



General Directorate of Scientific Research Center/ Salahaddin University-Erbil

Molecular Cloning & Recombinant Protein Lab.

Plan for academic year 2026-2027

Lab. Director

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1: Lab. Equipments & Techniques

International Standard

Status	Techniques	Item	Category	Equipment
	Enzyme Assay	Enzyme dependent substrate	Enzyme activity & properties	UV-visible double beam Spectrophotometer
	Recombinant Protein Production & Purification	Yeast Extract	Protein Production	Dry Bath
		Tryptone		Ice Maker
		D- (+)-Glucose		Incubator Shaker
		Agar		Sonicator
		Q5 High-Fidelity Master Mix		Hot Plate
		Gene JET Gel Extraction Kit		Vortex
		pET28a (+) vector		Concentrator
		E. coli Rosetta (DE3) pLysS		pH meter
		Kanamycin sulfate	Antibiotics	Refrigerated centrifuge
		IPTG	Inducers	Deep freezer -80
		NaH ₂ PO ₄	Buffer Components	Micro-centrifuge
		Na ₂ HPO ₄		Autoclave
		Potassium acetate		Bench top centrifuge
		Rubidium chloride		Water bath
		Manganese chloride		Set of Micropipettes
		Calcium chloride dihydrate		(SDS-PAGE) Protein analysis
		MOPS (free acid)		Protein gel casting stand & Gaskets
		Glycerol (MB grade)		SDS-PAGE system
		Sodium chloride		Power supply for electrophoresis
		Sodium deoxycholate		Mini see-saw rocker.
		Magnesium chloride hexahydrate		Nanodrop
		Magnesium sulfate heptahydrate		Detection System
		Potassium phosphate monobasic KH ₂ PO ₄		Analytical balance
		Potassium chloride KCl		Fridge 4 C°
		HiTrap Chelating HP	Chromatography	Freezer -20
		PD-10 Desalting Columns(8.3ml) resin	Culture Glassware	Set of Micropipettes
		Imidazole		Mini bench top incubator shaker
		NiSO ₄ ·6H ₂ O		Class II Biological Safety Cabinet (A2)
		Sterilin TM Borosilicate Glass Culture Tube		
		Corning® Erlenmeyer baffled cell culture flasks		
		Nalgene Sterile Sringe Filter, 02um,10-100ml	Filters	
✓ DONE	SDS-PAGE	Acrylamide/Bis-acrylamide	SDS-PAGE/Gel	Protein gel casting stand & Gaskets
		Tris-HCl Base	SDS-PAGE/Gel & Buffer	SDS-PAGE system
		SDS	SDS-PAGE/Gel & Buffer	Power supply for electrophoresis
		Ammonium Persulfate (APS)	SDS-PAGE/Gel	Dry-block
		TEMED	SDS-PAGE/Gel	Mini see-saw rocker.
		Sample Loading Buffer	Loading Buffer	Hot plate stirrer
		Glycine	SDS-PAGE/ Buffer	Sonicator
		Coomassie Brilliant Blue Stain	Gel Stain	Set of Micropipettes
		Methanol	Gel De-staining	
		Acetic acid	Gel De-staining	
		Protein Marker	Marker/Electrophoresis	
✓ DONE	Gene cloning	DNA Extraction kit	DNA extraction	Dry-block
		PCR Master Mix	Gene Cloning - PCR	Water bath
		PCR Primers		Top bench centrifuge
		DNase & RNase Free water		Vortex
		Agarose	Detection	Nanodrop
		TBE Buffer		PCR thermal cycler
		DNA Gel Stain (Safe dye)		Analytical balance
		DNA Ladder (100 bp)		Microwave
		Loading Dye		Agarose gel electrophoresis system
		Cloning vector (pET28a+)	Transformation	Power supply for electrophoresis
		Restriction Enzymes	Restriction Digestion	UV detection system
		DNA Ligase	Ligation	
		E. coli DH5α competent cells	Transformation	

Basic

2: Training course

Molecular Techniques Level 1	
1	Primer Design
2	DNA extraction
3	PCR
4	PCR Detection
Molecular Techniques Level 2	
1	Protein extraction
2	Gel preparation, Running, & staining
3	Gel scanning
Molecular Techniques (Advanced)	
Gene cloning	
1	PCR mediated cloning
2	Restriction digestion & Ligation
3	Transformation
4	Screening
Recombinant Protein Expression	
1	Transformation (Protein expression strain)
2	Colony Detection
3	Inoculation
4	Protein Over-expression
5	Protein Detection

3: Workshop

Speaker	
1	Multiple sequence alignment & Variant Call using Ugene
2	Data deposit in NCBI

4: post-graduate supervising

Stdudents		Degree
1	Zryan Ramazan Rasool	PhD
2	GhUrbat Mustafa Ahmed	MSc

5: Research

No.	Research title	status
1	Molecular Investigation of <i>ACE2</i> Collectrin Domain Polymorphisms in Gastrointestinal and Renal Functional Disorders	Research Article (ongoing)
2	Gene expression profiling in idiopathic granulomatous mastitis	Research Article (ongoing)
3	The Etiology of idiopathic granulomatous mastitis	Review article (ongoing)
4	<i>ACE2</i> Peptidase Domain Variants and Their Potential Impact on Renin–Angiotensin System-Related Diseases	Research Article (ongoing)
5	Human <i>ACE</i> and <i>ACE2</i> genes variants and COVID-19 disease in Iraqi Kurdistan region	Research Article (ongoing)
6	Genetic Analysis of <i>ACE2</i> Peptidase-Domain Exons 6 and 7 in Relation to Covid-19 Outcomes in a Kurdish Cohort	Research Article (ongoing)
7	The Role of <i>SLC6A4</i> (5-HTTLPR) Polymorphism in Moderating the Relationship Between Chronic Stress and Depression Risk	Research Article (ongoing)
8	Epigenetic Modifications of the <i>NR3C1</i> Gene in Individuals with Early-Life Trauma	Research Article (ongoing)
9	CRISPR-Cas9 Biosensors: From Genome Editing Technology to Next-Generation Diagnostic Platforms	Review article (ongoing)

Recombinant protein Application:

1. Medical & Pharmaceutical Applications

Therapeutic Proteins used directly as drugs:

- Insulin → diabetes
- Growth hormone → growth disorders
- Erythropoietin (EPO) → anemia
- Interferons → viral infections, cancer
- Clotting factors (VIII, IX) → hemophilia

2. Diagnostic Applications

- ELISA kits
- Rapid test kits
- Biosensors
- Diagnostic enzymes & antigens

3. Industrial Applications

- Detergents (proteases, lipases)
- Food industry (lactase, amylase)
- Biofuels (cellulases)
- Textile & leather processing

4. Agricultural Applications

- Recombinant plant growth regulators
- Pest resistance proteins (Bt toxins)
- Animal vaccines
- Feed enzymes (phytase)

5. Environmental & Biotechnological Uses

- Bioremediation enzymes
- Wastewater treatment enzymes
- Biosensors for pollutants
- Oil-spill cleanup enzymes